Status	Finished
Started	Wednesday, 21 October 2026, 11:38 PM
Completed	Wednesday, 21 October 2026, 11:41 PM
Duration	2 mins 50 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Merge Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Bubble Sort
- c. Quick Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **3**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is based on the divide-and-conquer approach
- b. It is less efficient than bubble sort
- c. It does not require recursion
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **4**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Being the most efficient sorting algorithm
- c. Using the divide-and-conquer approach
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **6**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes it easier to select items based on their relationship to the rest of the items
- b. It complicates the selection of items
- c. It makes the data unsorted
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are compared element by element
- c. They are discarded
- d. They are split again into smaller sublists

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It uses the heap data structure
- b. It recursively sorts subproblems
- c. It divides the list into sublists
- d. It bubbles up the largest element in each pass

The correct answer is: It bubbles up the largest element in each pass

Question **9**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes data harder to manage
- c. Makes code execution slower
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **11**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It requires external libraries
- b. It only works with integer arrays
- c. It sorts data out of the box efficiently
- d. It is less efficient than custom sorting algorithms

The correct answer is: It sorts data out of the box efficiently

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Quick Sort
- c. Bubble Sort
- d. Insertion Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Sorting data in a single pass
- c. Rearranging data without sorting
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **14**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Selection
- b. Searching
- c. All of the above
- d. Duplicates

The correct answer is: All of the above

Question **15**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Merging two datasets
- c. Removing duplicates from a list
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order